

An Exploration of Benford's Law

Mantissa Distributions, Lognormal Scaling, and Limit Behavior

A Monograph Adaption of Michael J. Sharpe (2006)

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Abstract

This paper presents a formal analytical investigation into Benford's Law, base- b mantissa distributions, and invariant probability measures on compact groups. By examining base-10 mantissas, fractional parts of logarithmic transforms, and lognormal data distributions as established by Michael J. Sharpe (2006) [1], we evaluate how multiplicative stochastic processes, triangular arrays, and normal approximations explain the origin of logarithmic first-digit phenomena. We establish the uniform asymptotic negligibility conditions required for mantissa convergence on S^1 , discuss applications in financial forensic auditing, and evaluate a companion interactive web simulation.

My Live Simulation Engine : [Access Interactive Simulation Web Application](#)

Core Math Domain: Scale Invariance & Goodness-of-Fit Tests

Primary Focus: Invariant Measure Mechanics on S^1 Mantissa Groups

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1 Introduction to Benford's Law and Mantissas

In empirical data collection across multi-scale physical and financial systems, the first significant digits $d \in \{1, 2, \dots, 9\}$ do not follow a discrete uniform distribution $\mathbb{P}(D = d) = 1/9$. Instead, smaller leading digits occur with significantly higher probability according to Benford's Law [1].

1.1 What is a Mantissa?

To formalize this phenomenon mathematically, we analyze the base- b mantissa of a positive real random variable $X > 0$ [1].

Mantissa $\text{man}(x)$: For any $x > 0$, x admits a unique base- b representation $x = \text{man}(x) \cdot b^n$, where $n = \lfloor \log_b x \rfloor \in \mathbb{Z}$ and the mantissa is constrained to $\text{man}(x) \in [1, b)$ [1].

Base Logarithm (Log): Taking the base- b logarithm relates the mantissa directly to the fractional part of the logarithm modulo 1 [1]:

$$\log_b \text{man}(x) = \{\log_b x\} = \log_b x - \lfloor \log_b x \rfloor$$

1.2 The Benford Density Function

A random variable X strictly obeys Benford's Law if the distribution of its logarithmic mantissa $\{\log_b X\}$ converges to a continuous uniform distribution $U(0, 1)$ on the unit interval $[0, 1)$ [1].

$$r(x) := \frac{1}{\alpha x} 1_{[1, b)}(x), \quad \text{where } \alpha := \ln b \tag{1}$$

Equation (1) defines the canonical Benford density function $r(x)$ over the support $[1, b)$ [1]. Integrating $r(x)$ over $[1, d)$ yields the classical leading digit probability $\mathbb{P}(\text{Leading Digit} \leq d) = \log_b(d + 1) - \log_b(d)$ [1].

2 Lognormal Distributions and Mantissa Closeness

Multiplicative growth processes naturally yield lognormally distributed values, providing a rich class of continuous distributions that conform closely to Benford's Law [1].

2.1 Defining Lognormal Data

A random variable X follows a lognormal distribution with shape parameter $\sigma > 0$ if $X = ke^{\sigma Z}$, where $k > 0$ is a scale constant and $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable [1].

Evaluating the fractional logarithmic part yields a linear transformation of Z [1]:

$$Y = sZ + m, \quad \text{where } s = \frac{\sigma}{\alpha}, \quad m = \log_b k \quad (2)$$

Equation (2) scales the standard normal variable Z by effective shape factor $s = \sigma/\ln b$ and shifts it by log-constant m [1].

2.2 Measuring Deviation from Uniformity

Michael J. Sharpe established explicit total variation and uniform metric bounds quantifying the distance between the fractional distribution $Y^0 = \{Y\}$ and a true uniform distribution [1].

$$\sup_{0 \leq y < 1} |F_{Y^0}(y) - y| \leq h(s) \quad (3)$$

Equation (3) establishes the supremum distance bound relative to uniform CDF $F(y) = y$ [1]. The error controlling function $h(s)$ decays exponentially with variance [1].

$$h(s) := \frac{\exp(-2\pi^2 s^2)}{\pi} + \frac{\exp(-9\pi^2 s^2/2)}{9\pi^3 s^2} \quad (4)$$

Equation (4) provides the precise evaluation of $h(s)$ [1]. As effective shape parameter $s \rightarrow \infty$, $h(s) \rightarrow 0$ rapidly, proving that lognormal variables with large variances strictly converge to Benford uniformity [1].

3 Triangular Arrays, Limit Theorems, and Group Invariance

To establish universal convergence for dependent or non-identically distributed sequence sums—such as financial log-return streams—we analyze triangular arrays of independent random variables [1].

3.1 Triangular Array Setup and Convergence Conditions

Let $\mathbb{F} := (X_{n,j})$ be a triangular array of independent random variables within each row $1 \leq j \leq k_n$, and let $S_n = \sum_{j=1}^{k_n} X_{n,j}$ denote the row partial sums [1].

Let $J_n \subset \{1, \dots, k_n\}$ select a primary variance-contributing subset [1].

$$\lim_{n \rightarrow \infty} \max_{1 \leq j \leq j_n} \sigma_{n,j}^2 \geq \sigma^2 > 0 \quad (5)$$

Equation (5) establishes the lower variance bound, ensuring that total row variance does not vanish as $n \rightarrow \infty$ [1].

$$\lim_{n \rightarrow \infty} \sum_{j=1}^{j_n} \mathbb{E} [X_{n,j}^2 \mathbf{1}_{\{|X_{n,j}| > \epsilon\}}] = 0 \quad \text{for every } \epsilon > 0 \quad (6)$$

Equation (6) enforces the classical Lindeberg condition for uniform asymptotic negligibility [1].

3.2 Convolution Decomposition and Invariance on S^1

Under conditions (5) and (6), the Central Limit Theorem guarantees that the partial sum S_n decomposes into independent components [1]:

$$S = X + Y \quad (7)$$

Equation (7) represents the asymptotic decomposition where $X \sim \mathcal{N}(\mu, \sigma_X^2)$ with $\sigma_X^2 \geq \sigma^2 > 0$, independent of Y [1].

Because the fractional operator $\{\cdot\}$ acts as a quotient map $\mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z} \cong S^1$ onto the circle group, the convolution of any continuous density with a normal density X of sufficient variance σ_X^2 wraps around S^1 uniformly. Thus, uniform asymptotic negligibility guarantees that the mantissa distribution $\{S_n\}$ converges to Haar measure uniformity on the compact group S^1 [1].

4 Applications and Practical Observations

The mathematical properties of mantissa convergence yield important practical applications in empirical data analytics:

- **Financial Market Auditing:** Multiplicative asset price processes and accounting ledgers exhibit lognormal or geometric Brownian behavior over long time horizons. Deviations from Benford mantissa uniformity signal potential data fabrication or ledger manipulation.
- **Empirical Lognormal Thresholds:** Lognormal datasets exhibit high fidelity to Benford's Law when shape parameter $\sigma > 1.2$ ($s > 0.52$), where bounding function $h(s) < 10^{-3}$ [1].
- **Multiplicative Aggregation:** Sums of log-transformed independent factors drive partial sums toward normal limits, causing mantissas of multi-factor multiplicative models to satisfy Benford's Law naturally [1].

5 Interactive Web Simulation Evaluation

To complement this theoretical monograph, an interactive single-file web application was evaluated: `benfords_law_sim.html`.

- **Live Interactive Application:** <https://benfordslawfinal.vercel.app/>
- **GitHub Source Code Repository:** <https://github.com/wheyyydant/math-drafts-and-si>
- **Engine Architecture:** Built as a standalone HTML/JS canvas dashboard capable of generating lognormal, uniform, and multi-factor stochastic datasets to evaluate empirical mantissa convergence against Benford bounds.

6 Concluding Remarks

We have presented an analytical formulation of Benford's Law and mantissa distributions based on Michael J. Sharpe's limit laws [1]. By evaluating lognormal error bounding functions $h(s)$ and triangular array decompositions under Lindeberg conditions, we demonstrate why multiplicative stochastic processes converge toward invariant Haar uniform measures on S^1 [1].

7 Mathematical Glossary & References

- Mantissa ($\text{man}(x)$):** The normalized fractional component of a number in base b , falling strictly within $[1, b)$ [1].
- Benford Density ($r(x)$):** The probability density function $1/(\ln(b) \cdot x)$ describing expected leading-digit frequencies [1].
- Circle Group (S^1):** The multiplicative group of complex numbers of unit modulus, isomorphic to $\mathbb{R}/\mathbb{Z}$.
- Lindeberg Condition:** A uniform asymptotic negligibility criterion for triangular arrays guaranteeing asymptotic normality of partial sums [1].
- [1] Reference:** Sharpe, Michael J. (2006). *Limit Laws and Mantissa Distributions*. Primary theoretical framework, bounding functions, and triangular array limit derivations [1].